

SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY

Faculty of Computing — Department of Software Engineering

Research Project — Periodic Log Entry Form

Project R26-SE-009 | Agent 2 | Ravindu Bandara Subasinha (IT22098450) | Reporting period 23.03.2026 – 25.08.2026

1. PROJECT AND STUDENT DETAILS

Project ID	R26-SE-009
Research Topic	Self-Optimized LLM Multi-Agent System for Software Engineering
Supervisor	Prof. Nuwan Kodagoda
Co-Supervisor	Mr. Eishan Weerasinghe
Student Name	Ravindu Bandara Subasinha
Student Registration No.	IT22098450
Individual Component	Agent 2 — Code Developer (AI Web Developer) and the Studio user interface
Reporting Period	23 March 2026 to 25 August 2026
Phase Covered	Requirement, implementation and integration phase, representing approximately 65% of the planned project work.

2. SUMMARY OF RP WORK CARRIED OUT DURING THE REPORTING PERIOD

This reporting period covers the development of Agent 2, the component that turns an approved specification into working software, together with the shared Studio interface through which the whole system is operated. The period began with a study of open-source project structures and prompt-driven generation approaches, from which the generation pipeline was designed: planning, component and page synthesis, artefact packaging, progress logging, a runtime sandbox for preview and a controlled repair path. That pipeline was implemented and given the Studio front end — workspace shell, dashboard, build timeline, file tree, code pane and preview pane. An assessment of the first version exposed the architectural weaknesses that drove a substantial rebuild: multi-stage architect and planner agents, a component registry with versioning and edit plans, centralised and tested error handling, route resolution with self-healing element-scoped patches, a template catalogue and a consolidated regression suite. The later part of the period covered split model routing, workspace and source-guidance tooling, module separation for maintainability, and the integration that brought the specification, QA and deployment components together with the builder behind one interface and one backend runtime.

3. DATE-WISE RP WORK (44 ENTRIES)

MARCH 2026

4 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
1	23.03.2026	Defined the scope of Agent 2 as the AI web developer that turns approved requirements into working software artefacts.
2	25.03.2026	Reviewed open-source project structures to identify reusable patterns for backend layout and frontend composition.

NO.	DATE	RP WORK CARRIED OUT
3	27.03.2026	Studied prompt-driven code generation approaches and recorded which of them produce reviewable, repeatable output.
4	31.03.2026	Designed the pipeline that converts a structured specification into component, page, service and project-level outputs.

APRIL 2026

8 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
5	02.04.2026	Planned generation support for reusable interface components, page templates and style variants.
6	06.04.2026	Designed the artefact manifest and the progress logging that make a generation run trackable and reviewable.
7	09.04.2026	Designed fallback behaviour so generation still produces usable output when the preferred model path is unavailable.
8	13.04.2026	Designed the repair strategy around controlled bug fixing rather than whole-project rewrites.
9	16.04.2026	Planned the runtime sandbox that previews generated React applications for validation and refinement.
10	20.04.2026	Implemented the build start, status and file endpoints together with background run tracking.
11	23.04.2026	Implemented component and page generation with style variants, catalogue endpoints and manifest packaging.
12	27.04.2026	Implemented sandbox planning, generation, refine and repair with a preloaded component library and preview wrapper.

MAY 2026

8 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
13	02.05.2026	Set up the Studio front-end base routes and the workspace shell.
14	05.05.2026	Reorganised the front-end structure and removed the superseded starter application.
15	08.05.2026	Built the dashboard cards, page flow and overall layout of the shared workspace.
16	12.05.2026	Added the build timeline and the progress states shown while a generation run is in flight.
17	15.05.2026	Built the shared builder components and wired the builder routes into the common navigation.
18	19.05.2026	Added the file tree, code pane and preview pane used to inspect a generated project.
19	22.05.2026	Completed responsive checks and visual polish across the builder screens.
20	27.05.2026	Reviewed the quality of the first version output and listed the architectural weaknesses to address.

JUNE 2026

9 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
21	01.06.2026	Added the architect agent that turns an approved specification into a concrete implementation plan.
22	03.06.2026	Introduced generation state management and the required-field rules applied to the architect prompt.
23	05.06.2026	Split the architect and planner into separate stages so high-level design is kept apart from file-level planning.
24	10.06.2026	Replaced the first prototype with a clean scaffold as the version 2 baseline.
25	15.06.2026	Centralised error handling and added unit tests so runtime failures never surface as unreadable objects.
26	17.06.2026	Implemented the route resolver and element-scoped patching with self-heal for targeted edits.
27	19.06.2026	Built the template catalogue and the version-on-success behaviour that protects a working build.
28	23.06.2026	Added the component registry, versioning and edit-plan support needed for controlled regeneration.
29	26.06.2026	Added backend regression tests covering the generation pipeline and the stability fixes.

JULY 2026

8 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
30	01.07.2026	Hardened component selection with a testable route-fallback library and a create resolver for data-driven pages.
31	06.07.2026	Implemented a cache-busted image and build flow and consolidated the stability test suite.
32	09.07.2026	Upgraded the planner so implementation plans cover file layout, dependencies and build order completely.
33	13.07.2026	Reskinned the Studio interface on a Tailwind foundation with a consistent set of theme tokens.
34	16.07.2026	Compared candidate models for each generation stage and recorded the differences in output quality.
35	20.07.2026	Implemented split model routing so every stage runs on the model best suited to it.
36	23.07.2026	Reorganised the application workspace and tightened the tracked-file rules for the desktop build.
37	28.07.2026	Divided the agent implementation into task-specific modules to improve maintainability.

AUGUST 2026

7 ENTRIES

NO.	DATE	RP WORK CARRIED OUT
38	03.08.2026	Added workspace tools and source guidance so the agent reuses existing project context when editing.

NO.	DATE	RP WORK CARRIED OUT
39	06.08.2026	Recorded a lessons store and added a browser reproducer for defects observed in generated applications.
40	11.08.2026	Tuned the generation prompts and added the console bug-fixer flow to the Studio.
41	14.08.2026	Added the theme, photo and image agent modules for asset-aware generation.
42	18.08.2026	Added project persistence and the builder scaffolding that carries saved project state.
43	21.08.2026	Split the agent exports into focused submodules and separated the tester-facing interface.
44	25.08.2026	Reviewed the integrated Studio across the builder, specification, QA and deployment panes ahead of consolidation.

4. PROJECT WORK PLANNED BEFORE THE NEXT EVALUATION PERIOD

- 1. Complete the version 2 integration so all four components run behind the single Studio interface and shared runtime without branch-specific behaviour.
- 2. Improve sandbox runtime debugging, preview reliability and automated repair quality for generated applications.
- 3. Extend the generated output to a fuller set of backend and frontend artefacts driven directly from the approved specification and plan.
- 4. Strengthen reuse so the agent learns from stored lessons and reference projects and transfers those patterns systematically.
- 5. Raise pre-handoff validation so builds reach Agent 3 with fewer structural and runtime defects, and complete the console bug-fixer loop.

5. DECLARATION AND SIGNATURES

I certify that the work recorded in this log book for the period 23 March 2026 to 25 August 2026 was carried out by me as the individual component contribution to research project R26-SE-009, and that the group-level contributions recorded here were made jointly with the other project members.

SUPERVISOR'S COMMENTS

Ravindu Bandara Subasinha STUDENT — IT22098450 Date: _____	Prof. Nuwan Kodagoda SUPERVISOR Date: _____	Mr. Eishan Weerasinghe CO-SUPERVISOR Date: _____
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